MIL–C–46941(MU)
1 AUGUST 1963

SUPERSEDING
FA–PD–MI–2384, REV. 2
4 JANUARY 1962

MILITARY SPECIFICATION

CARTRIDGE, THRUSTER, M146

1. SCOPE

1.1 This specification covers the Cartridge, Thruster, M146 for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL–A–2550 —Ammunition and Special Weapons, General Specification for.
MIL–E–5272 —Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL–I–45208 —Inspection Requirements, General Specification for.
MIL–T–46692 —Thruster, Cartridge Actuated, M16, Assembly.

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109 —Quality Assurance Terms and Definitions.
MIL–STD–639 —Surface Roughness Values Preferred for Army Ordnance Materiel.

DRAWINGS

MUNITIONS COMMAND

D8596426 —Cartridge, Thruster, M-146 Assembly.
C8830980 —Carton, Packing, Ammunition, for Cartridge, Thruster, M1-46.
C8830981 —Box, Packing, Ammunition, for Cartridge, Thruster, M146.
IEL–8596426 —Inspection Equipment List for Cartridge, Thruster, M146.

PUBLICATIONS

Frankford Arsenal Manual for Functional Tests—Cartridge Actuated Devices for Aircraft Ejection Systems.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Materials, parts, assemblies, chemical and physical requirements shall conform to the applicable drawings, referenced specifications and the following:

3.2 Initial production sample. Unless otherwise directed by the contracting officer, an

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:22Z
Check the source to verify that this is the current version before use.

initial production sample shall be required.

3.3 Protective finish. When protective finish is required by component details, finish requirements shall be as required on the drawing and referenced specification and shall apply to all surfaces of the assembled cartridge regardless of contour configuration.

3.4 Primer. The 72M Primer shall be used.

3.5 Cartridge assembly. Components shall be combined to produce an assembly which conforms to the requirements of Drawing D8596426.

3.5.1 *Vibration.* The assembled cartridge shall withstand vibration as prescribed in Specification MIL–E–5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.6 Functional requirements.

3.6.1 The cartridge shall not fail to fire.

3.6.2 *Powder charges.* The igniter and propellant charges shall be fixed by the contractor to produce the specified ballistic requirements. (See 6.1.)

3.6.3 *Functioning.* The cartridge when fired in an acceptable M16 Thruster shall produce sufficient pressure to comply with the functional requirements of MIL–T–46692.

3.7 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.7.1 *Metal defects.* All metal parts shall be free from cracks, splits, laminations, perforations, porosity, or any similar defect.

3.7.2 *Surface finish.* The requirements for surface finish are as detailed on the applicable drawing and referenced military standard.

3.7.3 *Burr.* No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.4 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, oil, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to Standard MIL–STD–109 to define quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with Specifications MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 Initial production sample. Prior to the initiation of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 30 cartridges and 10 sets of unassembled components including explosive components. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of Specification MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:22Z
Check the source to verify that this is the current version before use.

4.3.1 *Lot.*

4.3.1.1 *Presentation of lots.* The product shall be presented in accordance with Standard MIL–STD–105.

4.3.1.2 *Size of lot.* Unless otherwise specified by the procuring activity, an ammunition lot shall consist of 300 cartridges, maximum, exclusive of test samples and shall consist of:

(a) Components produced by homogeneous process.

(b) Propellant from not more than one lot.

(c) Igniter from not more than one lot.

(d) Primers from not more than one lot.

4.3.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and criteria in tables I and II. For an individual defect (single characteristic), the criteria in table I shall be used. For total defects within a major or minor group, the criteria in table II shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed in Standard MIL–STD–639 shall be used as a basis of comparison for surface roughness determination.

Table I

Lot size	Sample size	Acceptance numbers	
		Major	Minor
1— 300	100% Examination	0	0
301—1300	110	1	2
1301—3200	150	2	3

Table II

No. of defects in class	Total major defects allowable
2—3	1
4—7	2
8—19	3

No. of defects in class	Total minor defects allowable
2— 3	2
4— 7	3
8—16	4

4.3.2.1 *Classification of defects.*

4.3.2.1.1 *Cartridge, Thruster, M146, Assembly, Drawing D8596426.*

Defect	*Method of inspection*
Critical	
1. Depth to primer pocket face	Gage
2. Primer seating depth	Gage
3. Any part missing	X-Ray/Visual[1]
4. Improper assembly	X-Ray/Visual[1]
5. Mixed ammunition	Visual
6. Metal defective	Visual
7. Marking missing, incorrect or illegible	Visual
Major	
101. Cartridge profile failure	Gage
102. Diameter of body, max.	Gage
103. Diameter of head, max.	Gage
104. Length to datum diameter	Gage
105. Primer crimp improper	Visual
106. Protective finish inadequate or defective	Visual
107. Foreign matter	Visual
108. Burr	Visual
Minor	
201. Total length	Gage
202. Dent	Visual
203. Draw scratch	Visual
204. Fold, wrinkle, buckle or bulge	Visual
205. Varnish inadequate or missing	Visual
206. Surface finish improper	Visual

[1] The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the machine. The x-ray negatives, properly identified, shall be forwarded to the procuring agency.

4.3.2.1.2 *Primer, Percussion, 72M, Drawing C7645332, Prior to Assembly into a Cartridge.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:22Z
Check the source to verify that this is the current version before use.

Defect	Method of inspection
Critical	
1. Improper assembly	X-Ray[2]
2. Any part missing	X-Ray[2]
3. Mixed primers	Visual
4. Primer pellet broken, cracked or dislodged	Visual
5. Primer foil discolored	Visual
6. Metal defective	Visual
Major	
101. Burr	Visual
102. Foreign matter	Visual
Minor	
None defined.	

[2] Primer X-Ray. Prior to assembly into a cartridge, the primer shall be x-rayed with the side of the cup perpendicular to the rays to determine compliance with the requirements of the applicable drawing. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.1.3 *Carton, Packing, Ammunition, for Cartridge, Thruster, M146, Drawing C8830980.*

Critical
None defined.
Major
None defined.

Defect	Method of inspection
Minor	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.1.4 *Box, Packing, Ammunition, for Cartridge, Thruster, M146, Drawing C8830-981.*

Critical
None defined.
Major
None defined.

Defect	Method of inspection
Minor	
201. Contents loose	Manual/Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.2.2 *Primer examination.* Prior to assembly into a cartridge, each primer shall be visually examined for any discoloration of the foiling paper or malformation of any kind. Any discoloration of the foiling paper or malformation or failure to comply with the requirements of the drawing and referenced specification shall be cause for rejection of the primer.

4.3.3 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4, except as provided in Specification MIL–I–45208.

4.3.3.1 *Protective finish.* The number of units for protective finish tests shall be in accordance with the applicable specifications. Failure of any sample to comply with the requirements of the applicable specifications shall be cause for rejection of the lot.

4.3.3.2 *Airtightness.* Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the requirements of the drawing shall be cause for rejection of the cartridge.

4.3.3.3 *Vibration test, initial production sample.* Twelve of the 30 assembled, x-rayed cartridges of the initial production sample, properly identified, shall be subjected to vibration as directed in 3.5.1. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in [1] of 4.3.2.1.1. Evidence of any disintegration of the primer, propellant or igniter or derangement of any component shall be cause for rejection of the lot.

4.3.3.4 *Vibration test, production lot sample.* When it becomes necessary to make a change in the lot of primers, propellant, or igniter after production is initiated, 30 cartridges from the first production lot thereafter shall be tested in accodance with 4.3.3.3.

4.3.3.5 *Functioning.* The 30 test cartridges, including the 12 vibrated cartridges, of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:22Z
Check the source to verify that this is the current version before use.

initial production sample (or subsequent sample of 30 cartridges when a change in primers, propellant, or igniter has been made) and 14 cartridges from each subsequent lot shall be temperature conditioned and functionally tested. Failure of any cartridge to comply with the requirements of 3.6 shall be cause for rejection of the lot.

4.3.4 *Inspection equipment.* The examination and tests shall be made, using equipment listed on IEL–8596426, except as provided in Specification MIL–I–45208.

4.4 Test methods and procedures.

4.4.1 *Protective finish.* Test procedures for protective finish shall be in accordance with the applicable specification referenced on the drawing.

4.4.2 *Airtightness.* The method and procedure for determining airtightness shall be by use of equipment listed on IEL–8596426.

4.4.3 *Vibration.* The method and procedure for performing the vibration test shall be in accordance with Specification MIL–E–5272, Procedure XII.

4.4.4 *Functional tests.* The following tests will be performed by the Government:

4.4.4.1 *Initial production sample.* Cartridges of the initial production sample shall be grouped and temperature conditioned in accordance with 4.4.4.3 as follows: fifteen cartridges conditioned at minus 65 degrees Fahrenheit to minus 70 degrees F; 15 cartridges at plus 200 to plus 205 degrees F, and tested in accordance with 4.4.4.4. Each group of 15 cartridges shall include 6 of the 12 cartridges subjected to the vibration test required in 4.3.3.3.

4.4.4.2 *Production lot sample.* Cartridges of the production lot sample shall be grouped, temperature conditioned in accordance with 4.4.4.3 as follows: seven cartridges conditioned at minus 65 degrees F. to minus 70 degrees F, 7 cartridges at plus 200 degrees F. to plus 205 degrees F., and then tested in accordance with 4.4.4.4.

4.4.4.3 *Temperature conditioning.* Cartridges shall be loaded into acceptable M16 Thrusters with end caps to cover the gas ports during the conditioning period. The temperature of the propellant within the cartridge shall be considered the conditioning temperature and will be determined by means of a thermocouple attached to the center of the propellant charge of a specimen cartridge assembled in a standard test thruster device similar to the thrusters undergoing tests. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

4.4.4.4 *Test procedures.* Upon completion of the required temperature conditioning, the sample shall be ballistically tested in accordance with "Frankford Arsenal Manual for Function Tests—Cartridge Actuated Devices for Aircraft Ejection Systems" (or such method as is approved by the contracting officer) within 5 minutes after removal from the conditioning chamber. The sample shall be reconditioned if this time is exceeded. The total conditioning time, including any reconditioning at plus 200 degrees F. to plus 205 degrees F., shall not exceed 18 hours.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with Drawings C8830980 and C8830981.

5.2 Data cards. Data card information shall be in accordance with Specification MIL–A–2550.

6. NOTES

6.1 Advisory loading data:

Black Powder	(Class A–4)	1.5 grains plus or minus	0.10 grain
Propellant	(Class M2)	7.7 grains plus or minus	0.2 grain

6.1.1 *Advisory propellant grain dimensions:*

Length	0.046 inch
Diameter	0.038 inch
Perforation Diameter	0.003 inch
Web	0.015 inch
Number of Perforations	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:22Z
Check the source to verify that this is the current version before use.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—MU

Preparing activity:
Army—MU
Project No. 1375–A551

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:22Z
Check the source to verify that this is the current version before use.